Journal of Adhesion Science and Technology

ISSN: 0169-4243 (Print) 1568-5616 (Online) Journal homepage: www.tandfonline.com/journals/tast20

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To cite this article: Hui Yang & Zhengyu Xu (2024) Thermal fatigue life prediction of Ag-2.5Sn solder joint in flip chip, Journal of Adhesion Science and Technology, 38:18, 3525-3540, DOI: 10.1080/01694243.2024.2345955

To link to this article: https://doi.org/10.1080/01694243.2024.2345955

JOURNAL OF ADHESION SCIENCE AND TECHNOLOGY
2024, VOL. 38, NO. 18, 3525–3540
https://doi.org/10.1080/01694243.2024.2345955

Thermal fatigue life prediction of Ag-2.5Sn solder joint in flip chip

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ABSTRACT

The heat dissipation needs of high-power devices cannot now be met by traditional packing solders due to the fast development of microelectronics interconnect technology. In addition to having outstanding thermal and electrical conductivity, nano-silver-tin paste better satisfies the needs of electronic devices. Studying the thermal reliability of solder joints is extremely important, this article investigates the fatigue life of the nano-silver-tin solder joints in temperature cycling experiments. Based on the life test results, the thermal cycling fatigue life of the nano-silver-tin solder joint is determined to be 2880 cycles. Finite element software was used to replicate the chip's bonding process and assess the solder joint stress and strain distribution. The stress and strain of nano-silver-tin solder connections under thermal cycling load are accurately modeled using the Anand viscoplastic model and elastic model. The findings indicate that the contact region between the solder connections and copper columns is where the chip generates the most stress and strain. The fatigue life of the solder joint is estimated by using the Coffin–Manson modified equation, and this prediction aligns closely with the experimental results.

ARTICLE HISTORY
Received 9 September 2023
Revised 23 March 2024
Accepted 12 April 2024

KEYWORDS
Electronic package; anand constitutive model; fatigue life

1. Introduction

The soldering temperature of traditional tin-based solder is high, and the Si chip are acceptable in this case, but some polymer substrates cannot adapt to this temperature. So it is particularly important to find a substitute for traditional electronic packaging interconnect materials [1–3]. Nano silver slurry has a lower sintering temperature and is a substitute for traditional tin-based solder, although the interconnect joint can be obtained at a lower sintering temperature because the intermetallic composite is not created when the nano-silver slurry is sintered, the shear strength of the joint is poor so that cannot be applied to the chip packaging.

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Therefore, to increase the shear strength of the nano-silver paste, we need to add more low-melting alloying components. Indium and gallium have comparatively low melting points in doped alloys, but indium and gallium are highly expensive metals. Therefore, using the heterogeneous flocculation approach, we select nano-tin as the alloying component and create a nano-silver-tin slurry with a 2.5% tin (weight%) concentration. Tin is easily oxidized, thus a core-shell framework is employed to create a nano-silver-tin slurry because it can prevent tin oxidation and guarantee that tin is spread effectively throughout the matrix [4–7]. The research status of this core-shell material: Alayoglu team [8]once synthesized ruthenium nanoparticles by liquid phase experiment, and they also did a certain amount of experiments on the catalytic effect of the prepared core-shell material. Analysis and research show that when platinum coated ruthenium nano powder is used as the electrode material of hydrogen fuel cells, the prevention effect of CO poisoning is the best. Tao et al. [9] prepared Rh0.5Pd0.5 nanoparticles and Pt@Pd bimetallic powder was compared for its catalytic performance. In addition, they analyzed the different states in the preparation process by observing the various stages of the catalytic process. Tian Yanhong et al. [10] used a mild two-step liquid phase method to prepare copper silver core-shell nanoparticles, achieving the coating of small-sized silver nanoparticles with an average diameter of 6.9 nm on the surface of copper nanoparticles with an average diameter of 57.5 nm. The experimental results indicate that pulse current sintering of nano copper silver core-shell solder paste can quickly obtain a high-strength solder joint structure with high density and metallurgical connection on Cu–Cu substrate in a short period of time.

2. Preparation and fatigue experiments of nano-silver-tin paste

2.1. Preparation and sintering of nano-silver-tin paste

Because nano-silver has a higher isoelectric point and more positive charge under neutral conditions, while nano-tin has a more negative charge, we adopted heterogeneous flocculation, that is, the method of attracting opposite charges to each other, to make core-shell nano Sn@Ag composites. A benefit of the heterogeneity flocculation of the method is that it requires fewer resources and has relatively mild reaction conditions. To utilize this procedure, an appropriate surfactant need to be choosed, modify the solution's pH level, and measure the dimension of the compound particle.

In the aspect of surfactants, the relevant data were searched, and as compound surfactants, cationic cetyltrimethylammonium bromide (CTAB) and anionic sodium dodecyl sulfate (SDS) were used. Since both SDS and CTAB are soluble in ethanol, ethanol is chosen as the solvent. Since the diameter of tin and silver particles is nanoscale, the diameter of silver particles should be as small as possible, and agglomeration is easy to occur between particles when the size is too small, so the diameter of silver nanoparticles is determined to be 30 nm. The coating process can be aided by the fact that tin nanoparticles in the "inner" position often have a significantly bigger pore size than silver nanoparticles in the "outer" position. Consequently, it is found that the nano-tin particles have a diameter of 100 nm.

One technique used to manufacture the Sn@Ag core-shell material was heterogeneous flocculation. Firstly, the solution's pH value was found to be 7. Silver nanoparticles were adsorbed by anionic SDS to make their surface negatively charged. Similarly, the cationic CTAB absorbed the tin nanoparticles, leaving their surface positively charged.

Then, the nano-silver slurry is slowly added to the nano-tin slurry. As the nano-silver adsorbs the SDS surface with a negative charge, while the nano-tin adsorbs the CTAB with a positive charge, the nano-silver is slowly adsorbed to the nano-tin under the action of electrostatic attraction. With the increase of the nano-silver content, the electrostatic repulsion between colloidal particles is decreased, the thickness of the doubled electric layer is lowered, and then the bridging effect takes place and flocculation takes place. To make the nano Ag@Sn core-shell slurry, when nano-silver slurry is further added, the negative charge of nano-tin increases with the adsorption of nano-silver, and the charge goes up in negative value from zero, which repels the negatively charged nano-silver and causes the colloidal particles to disperse again.

The whole process is shown in Figure 1. Nano-silver-tin slurry with 2.5% Sn content was prepared by the above method.

Although the silver tin paste sintering process can also have three phases, it is different from the pure nano silver paste sintering process [11]. The first step was maintained at 80° C for 10 min, exceeding the solvolysis temperature to allow the ethanol to fully decompose. The second step was heated to a temperature of 210° C for 10 min, exceeding the decomposition temperature of the surfactant, so that the SDS was sufficiently decomposed. The last step is heated to 250° C, which is similar to the boiling point of the CTAB, so heat it to 250° C, and keep it warm for 30 min to ensure complete decomposition of the CTAB and full diffusion and reaction of tin and silver.

The sintering experiments of pure nano silver paste and nano Sn@Ag paste were carried out, and then the shear strength of the joint was measured by the shear instrument (DAGE 4000). As can be seen from Figure 2, the shear strength of Sn@Ag with a content of 2.5% is 50 MPa and that of pure silver is 40 MPa. Therefore, the shear strength of Sn@Ag nanoparticles is higher than that of silver nanoparticles [12].

2.2. Thermal cycling test of nano-silver and nano-silver-tin solder joints

Performance of nanosilver solder joints at cyclic temperature is one of the important reliability issues. The temperature cycling equipment used in the experiment is the

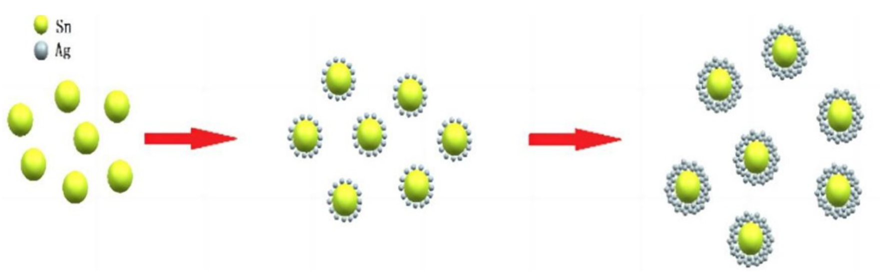

Figure 1. Heterogeneous flocculation process of silver-coated tin core-shell nanomaterials.

SD-308 single box temperature cycling test chamber, as shown in part 1 of Figure 3. To ensure that each cycle has the same cycle, this article uses a time control model to write programs, as shown in part 2 of Figure 3. Combined with JEDEC104 standard [13], the temperature cycle range is selected as −50 °C −150 °C, peak temperatures soaking time of 10 min. The temperature rise and fall time is set to 8 min, so each cycle is 36 min. The temperature change curve used in the experiment is shown in Figure 4.

The sample used in the experiment has a diameter of 100 μm Cu-pillar bumps flip chip, made using CB-600 inverted bonding machine. Based on the resistance

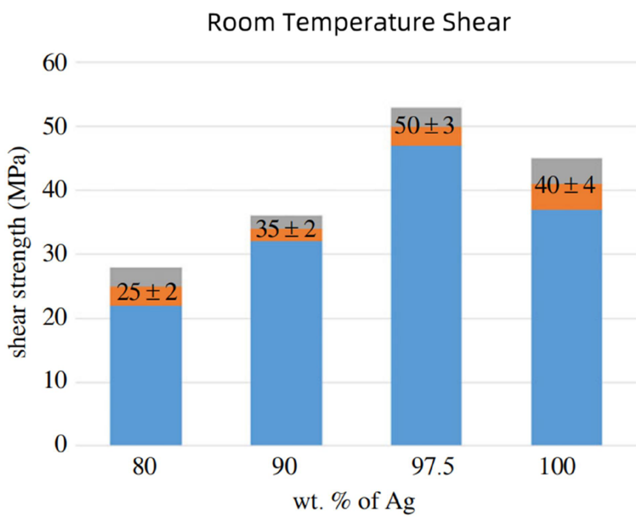

Figure 2. Shear strength diagram of sintered Ag + Sn joint (under 250 °C and 30 min of parameters).

Figure 3. Temperature cycling test chamber.

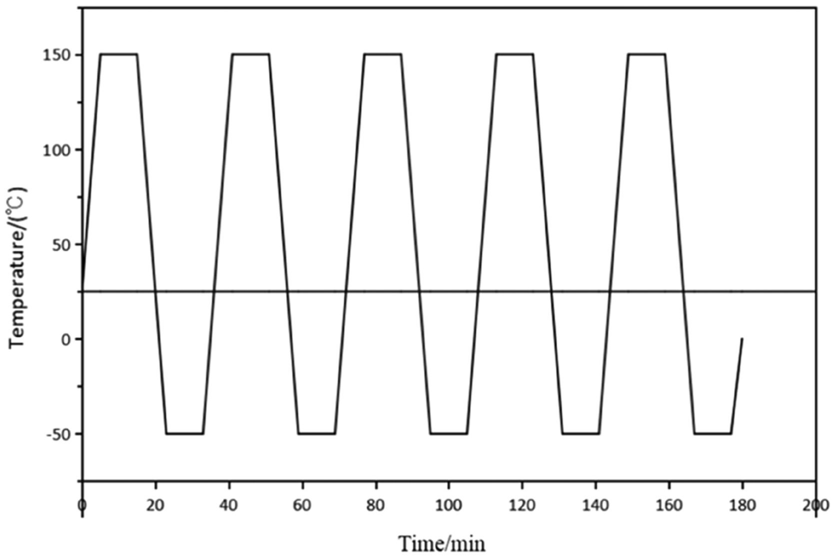

Figure 4. Temperature cycling curve for experimental setup.

monitoring results, the experimental chip takes out corresponding samples from the temperature cycling chamber when the resistance changes.

In the experiment, one of the daisy chains of the chip was selected for resistance testing to determine the fatigue life of internal solder joints. The resistance detection equipment is shown in part 3 of Figure 3, as a DC current source, it can output resistance changes during testing.

2.3. Resistance monitoring and failure analysis

Point A in Figure 5 represents the resistance diagram of the nano-silver solder joint, and point B in Figure 5 represents the resistance diagram of the nano-silver tin solder joint. The graphics of the two are consistent at the beginning, the resistance of the chip daisy chains changes with the change of cycle temperature, always fluctuating between 0.4 and 0.6Ω, and then the resistance delay rises. The nano silver solder joint has a complete open cycle failure at 2450 weeks, so the thermal fatigue life of the nano silver solder joint is 2450 weeks. The total open cycle failure occurs at 2880 weeks, so the thermal fatigue life of the nano-silver tin solder joint is 2880 weeks.

In Figure 6, the cross-sectional perspective displays the solder joint positioned at the array's corner exhibits a crack, where the resistance values are situated at point A. After undergoing 2450 cycles, a noticeable alteration in the resistance value of the sample becomes apparent. As illustrated in the figure, the solder joint positioned at the array's corner exhibits a crack, extending throughout the entire solder joint region. In Figure 7, the cross-sectional depiction of the chip at point B in the resistance variation diagram is presented. After 2880 cycles, the daisy chain of the sample has undergone complete disconnection, the solder joints located at the corners of the array exhibit substantial crack gaps. Five samples were tested and the

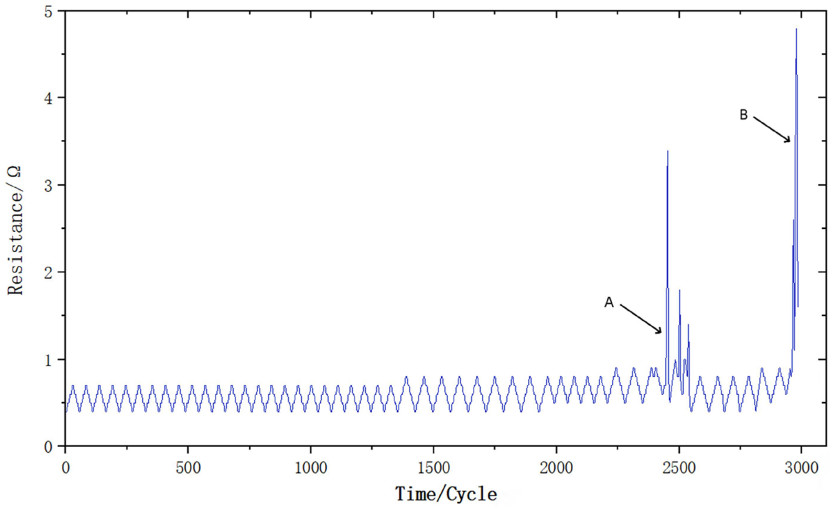

Figure 5. Resistance variation diagram of nano-silver and nano-silver-tin solder joints.

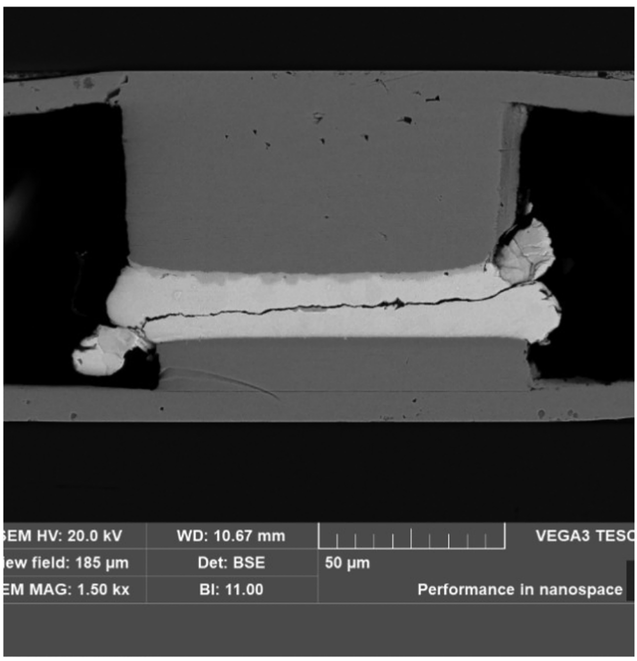

Figure 6. SEM image of the nano-silver solder joint.

results were consistent. Thus, the thermal fatigue life of nano-silver-tin solder joints is 2880 weeks, and that of nano-silver solder joints is 2450 weeks.

3. The constitutive equation of solder joint

In this article, the thermal reliability of nano-silver tin solder joints will be studied by using a finite-element modeling approach. Nano-silver-tin solder joints have a

Figure 7. SEM images of nano-silver-tin solder points.

low melting point, and the solder joints will be affected by elastic, and creep deformation during thermal cycling simulation. Therefore, we must take into account the physical characteristics of the solder junction, that is, to construct a constitutive equation that can accurately describe the mechanical behavior of the solder joint. Looking up relevant literature, because the elastic deformation of nano-silver-tin is small, but the viscoplastic deformation is obvious, Anand visco-plastic constitutive equation is adopted. There are two essential elements to the Anand model. First, any surface of the model can be used as a yield surface, and during distortion, loading and unloading requirements are not necessary. All stresses above zero result in plastic strain. Secondly, The inelastic impedance inside the material is represented by the model employing the change in impedance s as just one variable. The flow function of the Anand model is of hyperbolic sinusoidal type:

The relationship between deformation impedance and equivalent stress is

$$\sigma = c \cdot s; c < 1 \tag{3.1}$$

c- parameter, when the strain rate is a fixed value as follows:

$$c = \frac{1}{\xi} \sinh^{-1} \left\{ \left[\frac{\dot{\varepsilon}_p}{A} \exp\left(\frac{Q}{RT}\right) \right]^m \right\} \tag{3.2}$$

Among them: $\overset{*}{\varepsilon}_p$—inelastic strain rate;

ξ—stress multiplier; A—constant; Q—activation energy;

T—temperature; R—gas constant; M—strain rate sensitivity index.

Following the bending creep rule, the Anand constitutive equation is written as

$$\dot{\varepsilon}_p = A\left[\exp\left(-\frac{Q}{RT}\right)\right]\left[\sinh\left(\xi\frac{\sigma}{s}\right)\right]^{1/m} \tag{3.3}$$

The variable s can represent it as

$$s = h(\sigma, s, T)\dot{\varepsilon}_p \tag{3.4}$$

Among them, $h(\sigma, s, T)$—the Hardening function

The hardening function takes the form of

$$h = h_0\left|1-\frac{s}{s^*}\right|^{\alpha} sign\left(1-\frac{s}{s^*}\right) \tag{3.5}$$

Among them, h_0, α—Strain hardening parameters of materials; s^* is the saturation of s

From Equations 3.4 and 3.5:

$$s = \left[h_0\left(1-\frac{s}{s^*}\right) sign\left(1-\frac{s}{s^*}\right)\right]\dot{\varepsilon}_p \tag{3.6}$$

Among them

$$s^* = s'\left[\frac{\dot{\varepsilon}_p}{A}\exp\left(\frac{Q}{RT}\right)\right]^n \tag{3.7}$$

Among them s'—The saturation coefficient of s

n—Strain rate sensitivity;

s_0—The initial value of s

Joint Equation (3.1), (3.2), and (3.7), $\sigma^* = cs^*$

$$\sigma^* = \frac{s'}{\xi}\left[\frac{\dot{\varepsilon}_p}{A}\exp\left(\frac{Q}{RT}\right)\right]^n \sinh^{-1}\left\{\left[\frac{\dot{\varepsilon}_p}{A}\exp\left(\frac{Q}{RT}\right)\right]^m\right\} \tag{3.8}$$

Under isothermal conditions, the following formulas (3.1) and (3.6) can be obtained:

$$\frac{d\sigma}{d\dot{\varepsilon}_p} = ch_0 \left|1 - \frac{\sigma}{\sigma^*}\right|^{\alpha} sign\left(1 - \frac{\sigma}{\sigma^*}\right) \tag{3.9}$$

An integral of formula (3.9) yields an expression of stress:

$$\sigma = \sigma^* - \left\{ \left(\sigma^* - \sigma\right)^{(1-\alpha)} + (\alpha - 1)\left[\left(ch_0\right)\left(\sigma^*\right)^{-\alpha}\right]\dot{\varepsilon}_p \right\}^{\frac{1}{1-\alpha}}, \alpha \neq 1 \tag{3.10}$$

Among them $\sigma_0 = cs_0$,
Nine parameters of Anand equation: $A, \frac{Q}{R}, \xi, s', n, m, \alpha, h_0, s_0$, As shown in Table 1.
Their values can be determined by

1. The saturated stress can be obtained from the uniaxial tensile test of the constant strain rate and temperature.
2. The formula (3-8) is obtained by the nonlinear fitting method A, Q/R, n, m and s'/ξ;
3. The value of s'/ξ can be determined according to the formula (3-2). In the case of constant $c<1$, choose the appropriate ξ. Then determine the value of s'.
4. ch_0 and cs_0 are obtained by nonlinear fitting of the formula (3.10).
5. From the ξ value selected earlier, the c value can be determined, and therefore the values of h_0 and s_0 can be determined.

4. Modeling and simulation

4.1. Model

In the finite element simulation of the thermal cycle load, the model is treated as a 5-layer structure consisting of an upper substrate, UBM layer, nano-silver-tin solder, copper columns, and PCB. The chip is bonded to the substrate by nano-silver-tin paste, which contains 64 bumps with a diameter of 0.2 mm. To facilitate

Table 1. Anand model parameters of nano-silver tin [14].

Parameter	Value	Definition
A(s^{-1})	9.81	Pre-exponential coefficient factor
Q/R(J/mol)	5709	Activation Energy/gas constant
m	0.65720	Strain rate sensitive index of stress
n	0.00326	Strain rate sensitivity index of S *
ξ	11	Material constants
s' (MPa)	67.389	Coefficient of saturation value of deformation impedance
h_0 (MPa)	15800	Strain hardening constant
s_0 (MPa)	2.768	The initial value of the deformation impedance
α	1	Strain hardening index

calculation and reduce resource occupation, the following assumptions are made when establishing the model:

1. All solder joints in the array have the same geometric size and are evenly distributed.
2. Dense solder joint, no holes, pores, and other defects.
3. Since the chip is completely axisymmetric, the chip model is simplified, and the one-quarter structure of the chip containing the solder joint is adopted as the research object. The simplified model includes 16 solder joints in the 4×4 array layout, each spaced 400 μm apart. Figure 8 is the three-dimensional structure diagram of the model.

4.2. Define material properties

Ag-2.5Sn solder prepared by the heterogenous flocculation method has very hard Ag_3Sn particles. The silver matrix is primarily responsible for the plastic deformation of brazing metal. Therefore, in the definition of materials, except nano-silver paste is a viscoplastic material, the Anand model is used to describe the other materials are considered as elastic materials, material properties and temperature-independent. The properties of the materials are shown in Tables 2 and 3.

4.3. Grid division

Mesh generation is a key step before loading and computing. The grid generation process can affect the accuracy of simulation results. To make the effect of meshing

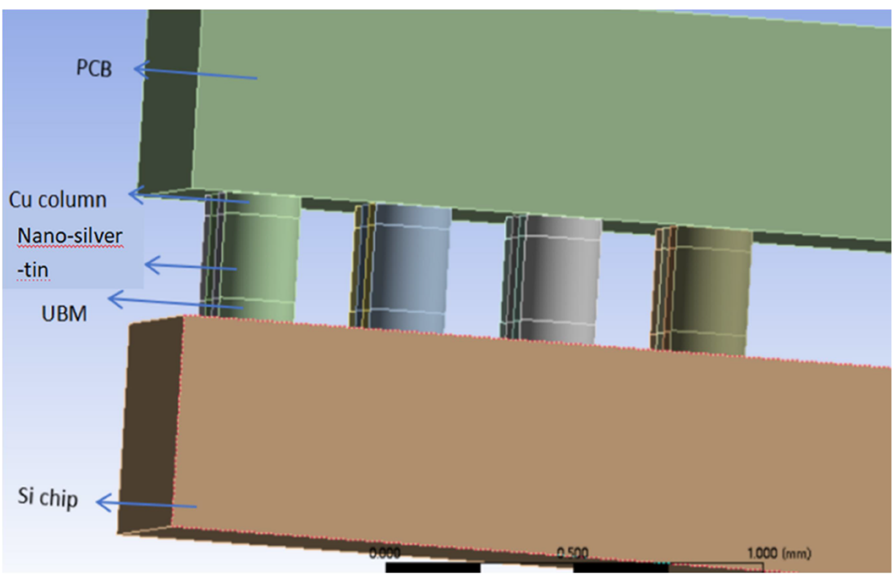

Figure 8. Simplified model annotation diagram.

Table 2. Material parameters of each component [15].

Materials	Dimensions (mm)	Modulus of elasticity (GPa)	Siméon Denis Poisson	Thermal expansion system (10^{-6}/°C)
Si chip	3.2×3.2×0.5	131	0.3	2.8
UBM layer	Φ0.2×0.05	100	0.3	18
Nano-silver paste	Φ0.2×0.2	Table 3	Table 3	19.6
Ag_3Sn	Φ0.1×0.08	80	0.35	18
copper column	Φ0.2×0.05	110	0.34	16.4
PCB	3.2×3.2×0.5	17.2	0.28	16

Table 3. Material parameters of nano-silver slurry at different temperatures [16].

Temperature	−50	0	25	60	120	150
Modulus of elasticity (GPa)	9.01	7.96	6.28	4.52	2.64	1.58
Poisson's ratio	0.29	0.32	0.37	0.39	0.41	0.45

meet the calculation precision and save the calculation time, the model is meshed by mapping grid, and the element is a hexahedron structure, for the part of nano-silver-tin, copper columns, and UBM layer, the size controller divides the length into 0.02 mm, and the chip and substrate part uses the size controller to divide the length into 0.1 mm, which divides 15420 grid units.

4.4. Apply load and boundary conditions

From previous literature, the stress and strain states of interconnect convex points fluctuate periodically during thermal cycle, and in most cases, after several cycles, the state tends to be smooth. The model calculates five cycles, and the temperature loading is carried out according to the requirements of Figure 4.

In general, the Si chip is connected to the PCB through the solder joint and is not fixed, while the PCB needs to be fixed to limit its deformation and displacement. So the PCB is set as a fixed surface, the temperature cycle load is applied to the entire model.

4.5. Results and Analysis

Figure 9 is the equivalent stress-strain diagram after the simulation. Figure 9 shows that the solder joint farthest from the center is the most susceptible to failure one. The reason is that the thermal expansion coefficient of each material is different. The substrate has a small deformation during the thermal cycling process, with a small deformation in the middle part and a large deformation in the edge part. As a result, the solder joint farthest from the center is subjected to greater stress and strain.

Figure 10 is the equivalent plastic strain diagram of corner solder joint. Where, green refers to the maximum plastic strain, red refers to the minimum plastic strain, and blue refers to the average plastic strain. From Figure 10, we can see that the plastic strain of the solder joints periodically accumulates with temperature.

The flimsy solder joint's stress-strain hysteresis curve is seen in Figure 11. From Figure 11, With the application of temperature load, the regularity of the curve changes, and the cumulative plastic deformation occurs inside the solder joints.

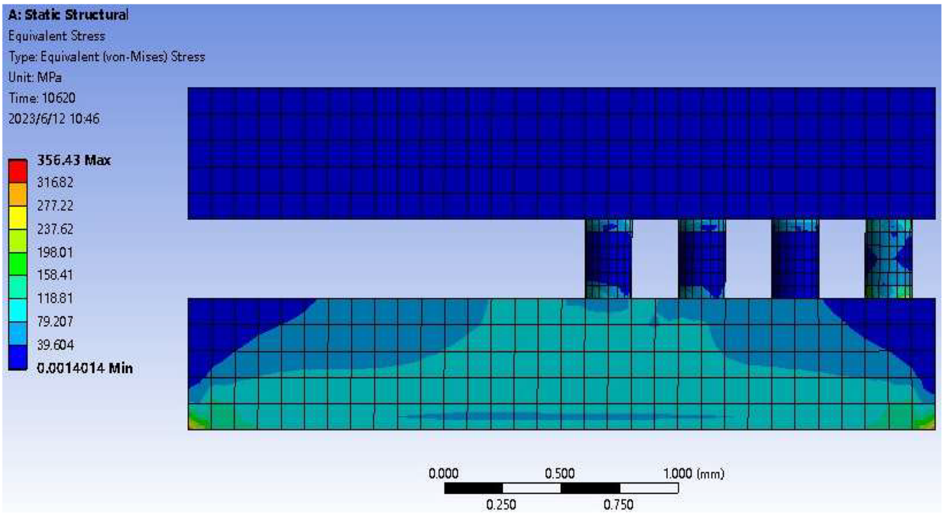

Figure 9. Chip equivalent stress diagram.

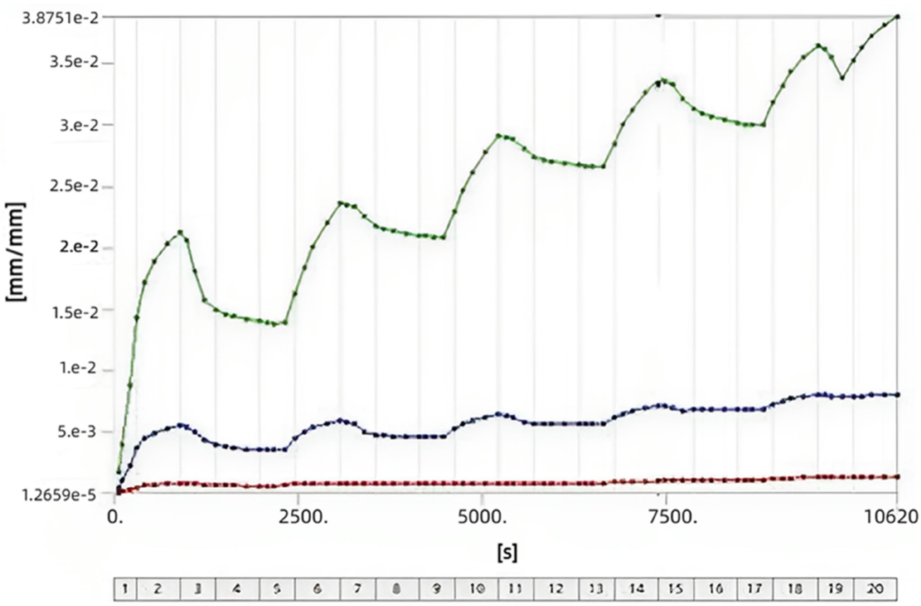

Figure 10. Relationship between equivalent plastic strain and time of corner solder joint.

5. Fatigue life prediction

In thermal fatigue simulation, the stress and strain of the solder joint are affected by many factors, such as temperature, solder material, solder joint structure, and size. Generally speaking, the nature of solder joint failure is that after a long time, due to the accumulation of plastic deformation, resulting in cracking or even failure [17, 18]. This is also evident in the simulation, the solder joint appears creep, a relaxation phenomenon, with viscoplastic characteristics. Therefore, considering the

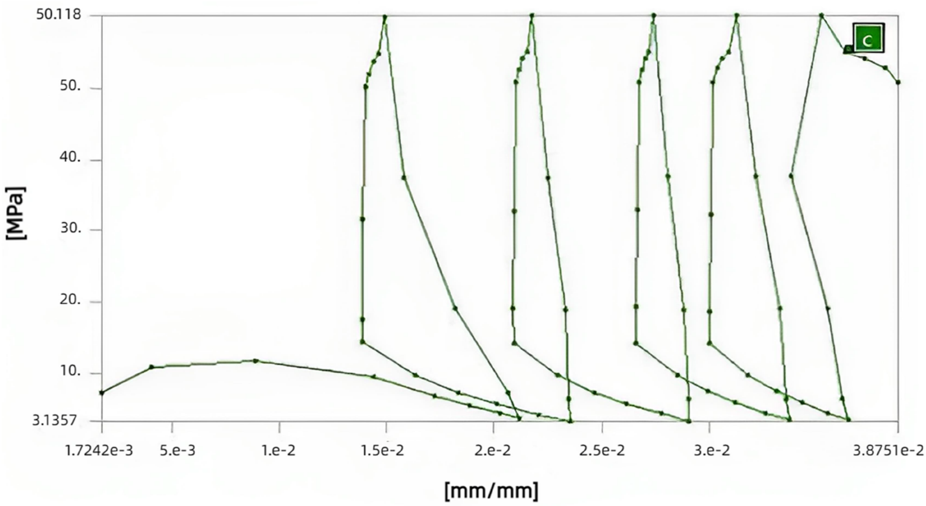

Figure 11. Internal node Stress-strain curve of corner solder joints.

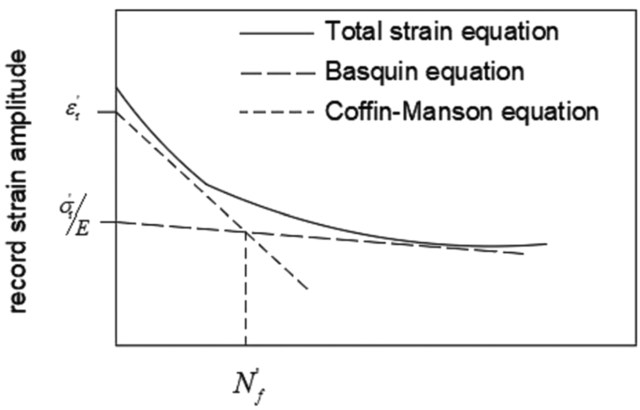

Figure 12. Relationship between total strain and life.

plastic deformation,we use the Coffin-Manson model to forecast the fatigue life of the solder joint.

The Coffin–Manson model links the amplitude of plastic deformation to the amount of fatigue fracture circulates. The relationship between the two is as follows:

$$N_f = C\left(\Delta\varepsilon_p\right)^{-\beta} \tag{5.1}$$

In the formula, N_f—The amount of fatigue fracture circulates;

C and β—material constants; $\Delta\varepsilon_p$—plastic strain range.

In practice, plastic deformation is the main influence on fatigue life, but elastic deformation also has a certain influence on fatigue damage. Therefore, Engelmaier modify the Coffin–Manson model. As shown in Figure 12, the model combines plastic and elastic influences to form a total strain curve, which is mainly controlled by Coffin-Manson in the front and the Basquin equation in the back.

The amended model is accurate enough to forecast the thermal fatigue life of the solder joint. However, this model only discusses the influence of deformation on the solder joint and does not consider the influence of temperature, which reduces the reliability of its life prediction. Therefore, the newly proposed Coffin-Manson amend model (also known as the Coffin-Manson low-cycle fatigue equation), which incorporates the effect of temperature on the solder joint, is one of the most accurate derived equations, as follows:

$$N_f = \frac{1}{2}\left(\frac{\Delta\gamma}{2\varepsilon_f'}\right)^{\frac{1}{c}} \tag{5.2}$$

Among them, $\Delta\gamma$ stands for equivalent plastic shear strain range $\Delta\gamma = \sqrt{3}\Delta\varepsilon$, ε_f' stands for fatigue ductility coefficient (This article take ε_f'=0.325).

$$C = 1.74\times10^{-2}\ln(1+f) - 6\times10^{-4}T_m - 0.442 \tag{5.3}$$

Among them, C—fatigue ductility exponent;

T_m—average temperature;

F—thermal cycle frequency.

The Coffin-Manson low-cycle fatigue equation not only makes the calculation process simple, but also the calculation result is more accurate, so this model is adopted in this paper.

Through the simulation results of finite element, the strain amplitude of each part is calculated and the maximum value is obtained. Since the equivalent plastic strain is accumulated periodically with the temperature, it is enough to choose a temperature cycle at random to calculate the equivalent plastic strain difference. Select the fourth temperature cycle from 6300s to 8460s for this node. As shown in Figure 11, the minimum and maximum equivalent plastic strain values are 0.0265 and 0.0337 respectively. The difference between the two is $\Delta\varepsilon$=0.0072, If you plug in a formula $\Delta\gamma = \sqrt{3}\Delta\varepsilon$, you get $\Delta\gamma$=0.01247, Then the average temperature is calculated T_m= (150-50)/2 = 50 °C, Temperature cycle time is 36 min, Therefore, the frequency of the temperature cycle f = 1.67cyc/h is obtained. Substituting the T_m and f values calculated above into the formula (5-3), we get C=−0.455, After entering the computed values for C, ε_f', and $\Delta\gamma$ into the formula (5-2), the result is N_f= 2,969 cycles. Similarly, it can be calculated that the fatigue life of nano silver solder joints is 2655 cycles, it can be seen that nano silver coated tin not only improves its strength, but also has a longer fatigue life compared to nano silver solder joints. This prediction aligns closely with the experimental results, proving the credibility of the simulation results.

6. Summary

This article investigates the fatigue life of the nano-silver-tin solder joints in temperature cycling experiments. Based on the life test results, the thermal cycling fatigue life of the nano-silver-tin solder joint is determined to be 2880 cycles. The chip adopts Anand viscoplastic plus elastic model structure, which is simulated by 3D software modeling and finite element method. It is found that the strain of nano-silver tin is the largest at the outermost solder joint, and the most fragile position of the solder joint is determined. The simulation can offer some help to the finite element analysis of other metal doped nano-silver slurry. Although the equivalent stress of the solder joint is fundamentally constant, the plastic strain will cause a rise over time that will cause the solder joint to fail. The fatigue life of the solder joint of nano-silver-tin paste is estimated by using the modified Coffin-Manson equation. It was found that nano-silver-tin solder paste has a longer service life and higher reliability compared to nano-silver solder joints.

Acknowledgment

Thanks to You Huan from Guilin University of Technology for your help in writing and reviewing this paper.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability statement

The data of this study are available from the corresponding author upon reasonable request.

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